

There is another redundancy: when an organ can be employed to perform a certain function that is not its current central one. My friend Peter Bevelin links this idea to the “spandrels of San Marco,” after an essay by Steven Jay Gould. There, the necessary space between arches in the Venetian cathedral of San Marco has led to art that is now central to our aesthetic experience while visiting the place. In what is now called the *spandrel effect*, an auxiliary offshoot of a certain adaptation leads to a new function. I can also see the adaptation as having a dormant potential function that could wake up in the right environment.

The best way to illustrate such redundancy is with an aspect of the life story of the colorful philosopher of science Paul Feyerabend. Feyerabend was permanently impotent from a war injury, yet he married four times, and was a womanizer to the point of leaving a trail of devastated boyfriends and husbands whose partners he snatched, and an equally long one of broken hearts, including those of many of his students (in his day, certain privileges were allowed to professors, particularly flamboyant professors of philosophy). This was a particular achievement given his impotence. So there were other parts of the body that came to satisfy whatever it was that made women attached to him.

Mother Nature initially created the mouth to eat, perhaps to breathe, perhaps for some other function linked to the existence of the tongue. Then new functions emerged that were most probably not part of the initial plan. Some people use the mouth and tongue to kiss, or to do something more involved to which Feyerabend allegedly had recourse.

Over the past three years I became obsessed with the notion that, under epistemic limitations—some opacity concerning the future—progress (and survival) cannot take place without one of these types of redundancy. You don’t know today what may be needed tomorrow. This conflicts very sharply with the notion of teleological design we all got from reading Aristotle, which has shaped medieval Arabic-western thought. For Aristotle, an object had a clear purpose set by its designer. An eye was there to see, a nose to smell. This is a rationalistic argument, another manifestation of what I call Platonicity. Yet anything that has a secondary use, and one you did not pay for, will present an extra opportunity should a heretofore unknown application emerge or a new environment appear. The organism with the largest number of secondary uses is the one that will gain the most from environmental randomness and epistemic opacity!

Take aspirin. Forty years ago, aspirin’s *raison d’être* was its antipyretic (fever-reducing) effect. Later it was used for its analgesic (pain-reducing) effect. It has also been used for its anti-inflammatory